DataNova: Automated EDA Report

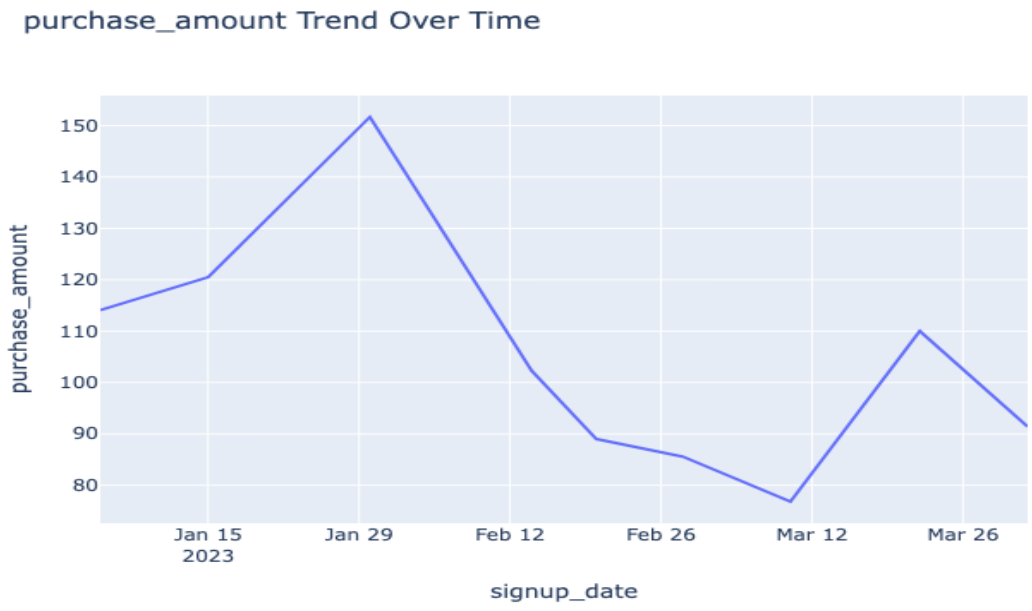
Format: Executive | Domain Detected: Sales

Executive / Dataset Overview - The dataset captures 10 transactions across 6 core fields (customer ID, age, income, signup date, region, purchase amount). - It represents a snapshot of sales activity, linking demographic attributes to monetary outcomes. - The small sample size limits statistical confidence but is sufficient for a quick health check and to surface immediate data-quality concerns. - Key business variables-*income* and *purchase_amount*-are present, enabling early exploration of revenue drivers.

Data Quality & Profiling Insights - **Missing data:** Only the *signup_date* field has a gap (1 record, 10 % missing); this could affect cohort analyses that rely on tenure. - **Outlier detection:** One *income* value (71,000) exceeds the calculated upper bound (70,375), suggesting a potential entry error or a high-value customer worth separate review. - **Schema consistency:** All columns conform to expected data types; no dtype mismatches were detected. - **Sample size:** With only 10 rows, any pattern may be driven by individual records; larger data collection is needed for robust modeling.

Strategic / Operational Takeaways - **Data hygiene priority:** Resolve the missing *signup_date* and verify the outlier income record before building predictive models or reporting on customer lifetime value. - **Enrich the dataset:** Augment with additional transactions, product details, and interaction timestamps to enable segmentation and trend analysis. - **Quick win opportunity:** Even with limited data, a preliminary correlation check between *income* and *purchase_amount* can identify high-spending segments for targeted marketing. - **Governance recommendation:** Implement validation rules at data-entry (e.g., income range checks) to prevent future outliers and ensure completeness of critical fields. - **Decision-making impact:** Clean, complete data will improve the accuracy of revenue forecasts, customer acquisition cost calculations, and ROI assessments for regional campaigns.

Visual Analysis




Top region by purchase_amount
purchase_amount
300
250
200
150
100
50
0
North
South
East
West
region